Training Program for Pyspark Support for SQL on Big Data

Bridge Role — Supporting Until Permanent Position

Structured, Intensive, Career-Level Training

This is a bridge role intended to support you until you secure a permanent position. The training path is structured and suggestive. Realistically, it takes 6 months to 2 years to gain full readiness for the role, depending on

Company QcFinance.In
Implementation for US partner NJ USA

Additional Learning Resources:
In addition to the Udemy courses listed below, there are **YouTube playlists** based on the project that you must follow. These will be shared incrementally as you progress through each phase.

The courses must be taken in the order listed below, as each builds on the previous one. Training will focus heavily on **Python**, **py spark code edits in distributed systems**, **testing and re-running pyspark code**, and **understanding HIVE Joins in big-data setup**.

Let me know once you start, and we will review progress weekly.

Your role might not be exactly what is in the course, but it will give you some idea of what skills are needed and how to approach them. As time goes by, the role might evolve and become more focused on Copilot and AI-based prompts and agents. However, the working style, methodology, and culture will remain the same.

Best regards,
Shivgan Joshi India MP Indore

Complete Course Links (Suggested Sequence)

Course 1: Note taking Course and Shadow Analyst for C Suite Analyst
https://www.udemy.com/course/note-taking-course-and-shadow-analyst-for-c-suite-analyst/

Course 2: Remote & Independent Working in Managerless Environments
https://www.udemy.com/course/remote-independent-working-in-a-managerless-environment-101/

Course 3: Connecting & Managing Ubuntu Bridge Computers for Remote Work
https://www.udemy.com/course/connecting-managing-ubuntu-bridge-computers-for-remote-work/

Course 4: Debug & Fix LaTeX Errors in Enterprise Environments (NEW)
https://www.udemy.com/course/debug-and-fix-latex-errors-in-enterprise-environments/

Course 5: Enterprise Data Permissions, Sources, Tools & Connections (NEW)
https://www.udemy.com/course/enterprise-data-permissions-sources-tools-connections/

Course 6: Research & Writing Skills for Essays in Controlled Environment on Bridge Node
https://www.udemy.com/course/research-writing-skills-for-essays-in-controlled-environment

Below are Job specific, check with your project ————————————————-

Course 7: Big Data Infrastructure Administration — Ticket-Based Learning
https://www.udemy.com/course/big-data-infra-admin-101-ticket-based-learning/

Course 8: Python Full-Stack Computational Framework (Beginner)
https://www.udemy.com/course/python-full-stack-computational-framework-for-begineers-101/

Course 9: Full-Spectrum Comprehensive Testing & Error Reading
https://www.udemy.com/course/full-spectrum-comprehensive-testing-error-reading-101/

Course 10: Python Big Data Refresher — Intermediate (Quiz Based)
https://www.udemy.com/course/refresher-python-big-data-intermediate-quiz-based-course-102/

Course 11: Python Full-Stack Backend & ML Engines
https://www.udemy.com/course/python-full-stack-and-backend-engines-for-mc-ml-engines-102/

Course 12: Writing Tests for Simulation Engineering & Code Conversion
https://www.udemy.com/course/writing-tests-for-simeng-python-code-conversion-concepts-101/

Course 13: Full-Stack Computational Framework for Big Data Simulations
https://www.udemy.com/course/full-stack-computational-framework-for-big-data-simulations/

Course 14: Big Data Code Changes for Full-Stack Simulation Engine 2026
https://www.udemy.com/course/big-data-code-changes-for-full-stack-simulation-engine-2026/

Course 15: Converting Standalone Python Code to Distributed Analytics Code
https://www.udemy.com/course/converting-standalone-code-to-distributed-code-for-analytics/

Course 16: FinMod Analytics with Min Guidance, No Info, Uncertain Env (NEW)
https://www.udemy.com/course/finmod-analytics-with-min-guidance-no-info-uncertain-env/

Training Assumptions

- Each course requires approximately 2–4 days
- Average effort: 4–6 hours per day
- Training cadence: one/two course per week. Get bi-weekly feedback from the instructor assigned to you and get your essays evaluated
- Initial readiness: approximately 10-14 weeks (all 16 courses)
- Professional mastery requires extended real-world practice and hence this is proctored program, with calls with your instructor 2-3 times a week
- Ideally training should be done on Ubuntu edge Node

Part 2: Detailed Training Plan (Step-by-Step, Time-Based) Draft version prepared for Last role

Assumptions

- Each course takes 2–4 days
- Average effort: 4–6 hours/day
- Training pace: 1 course per week
- Total timeline below focuses on initial readiness (approx. 16 weeks)
- Full mastery continues beyond this with practice

Phase 1: Foundation, Shadow Analyst & Bridge Setup (Weeks 1–4)

Week	Course	Time	Goal
1	Course 1: Note taking & Shadow Analyst for C Suite	2–3 days	Learn executive-level note taking, shadowing techniques, and how to support C-suite decision-making with structured observations
2	Course 2: Remote & Independent Working	2–3 days	Learn how to work independently; understand ticket-based and manager-less environments; set expectations for self-driven technical work
3	Course 3: Ubuntu Bridge Computers for Remote Work	3–4 days	Set up and manage Ubuntu-based bridge machines; understand networking, SSH, and remote connectivity essential for distributed infrastructure
4	Course 4: Debug & Fix LaTeX Errors (NEW)	2–3 days	Master LaTeX error resolution in enterprise docs; essential for simulation reports, technical papers, and controlled-environment essays

Phase 2: Data Permissions, Research Writing & Infrastructure (Weeks 5–8)

Week	Course	Time	Goal
5	Course 5: Enterprise Data Permissions & Tools (NEW)	3–4 days	Understand data governance, access control, source authentication, and enterprise connectivity — critical for real big-data environments
6	Course 6: Research & Writing in Controlled Environment	2–3 days	Structured documentation, clear technical writing, and research skills for controlled big-data environments
7	Course 7: Big Data Infrastructure & Ticket-Based Learning	3–4 days	Understand big-data systems; learn production-style workflows; exposure to infrastructure issues and troubleshooting
8	Course 8: Python Full-Stack Computational Framework (Beginner)	3–4 days	Python fundamentals; data structures and computational thinking; writing clean, reusable Python code

Phase 3: Testing, Error Reading & Python Intermediate (Weeks 9–12)

Week	Course	Time	Goal
9	Course 9: Comprehensive Testing & Error Reading	2–3 days	Debugging skills; understanding stack traces; reading and fixing errors independently
10	Course 10: Python Big Data Refresher (Intermediate)	3–4 days	Reinforce Python concepts; apply Python to data-heavy problems; build confidence in writing intermediate-level code
11	Course 11: Python Backend & ML Engines	3–4 days	Backend computation; understanding Python in ML / analytics pipelines; performance-oriented programming
12	Course 12: Writing Tests for Simulation & Code Conversion	2–3 days	Writing tests for complex logic; ensuring correctness during refactoring; quality control mindset

Phase 4: Distributed Systems, Simulation Engine & FinMod Analytics (Weeks 13–16)

Week	Course	Time	Goal
13	Course 13: Full-Stack Computational Framework for Big Data	3–4 days	End-to-end computation pipelines; simulation-based design; scaling logic from small to large systems
14	Course 14: Big Data Code Changes for Full-Stack Simulation Engine 2026	3–4 days	Hands-on modifying simulation engine code for big-data scale; real-world code change management
15	Course 15: Converting Standalone Code to Distributed Code	3–4 days	Distributed computing concepts; parallel execution; production-grade analytics workflows — capstone for distributed PySpark readiness
16	Course 16: FinMod Analytics (NEW)	3–4 days	Financial modeling under minimal guidance and uncertainty; prepares you for real-world ambiguous requirements and high-stakes decision support

Summary Timeline (Updated)

Phase	**Duration**
Foundation, Shadow Analyst, Remote Work, Ubuntu Bridge & LaTeX Debugging	4 weeks
Data Permissions, Research Writing & Infrastructure	4 weeks
Testing, Error Reading & Python Intermediate	4 weeks
Distributed Systems, Simulation Code, FinMod & Scale	4 weeks
Initial Readiness	**~16 weeks**
Professional Mastery	6–24 months with practice

How to Use This Plan

- Follow strict order
- Do hands-on coding alongside videos
- Re-do exercises if unclear
- Expect difficulty — that's normal
- This is career-level training, not casual learning

Appendix: Expanded Acronyms & Pedagogical Basis

Expanded Acronyms (Full Forms)

- **ML** — Machine Learning
- **MC** — Monte Carlo (simulation methods)
- **SimEng** — Simulation Engineering
- **Infra** — Infrastructure
- **Admin** — Administration
- **Python** — Not an acronym; a programming language named after Monty Python
- **API** — Application Programming Interface
- **CI/CD** — Continuous Integration / Continuous Deployment
- **SSH** — Secure Shell
- **Ubuntu Bridge** — A computer acting as a network bridge/gateway for remote distributed systems
- **LaTeX** — Document preparation system for high-quality typesetting
- **FinMod** — Financial Modeling

Why This Strict Order? (Pedagogical Explanation — Updated for 16 Courses)

The training plan follows a **cumulative skill ladder**.

1. **Week 1 (Shadow Analyst)** — Executive communication and observation skills; foundation for understanding business context.
2. **Week 2 (Remote Work)** — Metacognitive foundation: how to learn independently.
3. **Week 3 (Ubuntu Bridge Computers)** — Before any cloud or big-data infra, you must know how to set up and manage the actual bridge machine (Ubuntu) that connects remote work environments. Essential for the "bridge role."
4. **Week 4 (LaTeX Debugging — NEW)** — Enterprise documentation is written in LaTeX. Errors in simulation reports block progress. Debugging LaTeX is a core skill for controlled-environment essays and technical communication.
5. **Week 5 (Enterprise Data Permissions & Tools — NEW)** — Real big-data environments enforce strict access control. Understanding permissions, data sources, and enterprise connectors prevents security violations and ensures compliant workflows.
6. **Week 6 (Research & Writing)** — Structured documentation and controlled-environment research skills. Critical for tickets and simulation reports.

7. **Week 7 (Big Data Infrastructure)** — Understand ecosystems (clusters, pipelines) before writing code.
8. **Week 8 (Python Beginner)** — Core programming fundamentals.
9. **Week 9 (Testing & Error Reading)** — Debugging taught early for self-correction.
10. **Week 10 (Python Intermediate Refresher)** — Reinforcement through quiz-based, data-heavy problems.
11. **Week 11 (Backend & ML Engines)** — Performance-oriented Python for real pipelines.
12. **Week 12 (Writing Tests)** — Non-negotiable for distributed systems.
13. **Week 13 (Full-Stack Big Data Framework)** — Integrates all previous skills.
14. **Week 14 (Big Data Code Changes for Simulation Engine)** — Real-world modification of existing simulation engine code. Teaches how to safely introduce big-data changes without breaking simulations.
15. **Week 15 (Standalone → Distributed)** — Capstone: convert single-machine code to parallel, distributed execution.
16. **Week 16 (FinMod Analytics — NEW)** — Financial modeling with minimal guidance and uncertain information. Trains you to operate effectively in ambiguous, high-stakes environments — exactly the conditions of real enterprise support.

Cognitive Load Management

- Early weeks focus on *one new major concept* per week.
- Shadow analyst skills establish business communication before technical depth.
- Ubuntu bridge skills are introduced immediately after remote work — practical before theoretical infrastructure.
- LaTeX debugging is placed before research writing so documentation errors can be resolved independently.
- Enterprise data permissions come before infrastructure to build a security-aware mindset.
- Research/writing is placed after testing to reinforce documentation of errors and test results.
- Big-data code changes come before final conversion to distributed code — you first learn to modify existing engines, then redesign from standalone to distributed.
- FinMod Analytics is the final course because it simulates real-world ambiguity and minimal guidance — the ultimate test of self-sufficiency.

Time-to-Competence Realism (Updated for 16 Courses)

- **16 weeks** → Initial readiness (can complete supervised tickets, understands bridge role basics, handles enterprise documentation and permissions)
- **6 months** → Consistent independent work on medium-complexity tasks
- **1–2 years** → Full mastery (architecting distributed solutions, mentoring others)

Document generated from the original training plan by Shivgan Joshi. All Udemy links are preserved as provided. Updated with all 16 courses in strict sequential order, including three new courses: LaTeX Debugging, Enterprise Data Permissions, and FinMod Analytics.